

ENGINEERING DESIGN REPORT

AquaSense PCB Specifications & Verification

Automatic Touchless Water Dispenser

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This report documents the AquaSense PCB architecture, power-supply arrangement, grounding strategy, design software, controller integration, and recorded KiCad verification results.

1. Board Architecture and Software

The AquaSense control board is specified as a two-layer printed circuit board. The front layer is used primarily for signal routing and component interconnections, while the back layer incorporates a copper pour for the common ground network. This arrangement supports compact routing and provides a broad return path for the connected electronic circuits.

The PCB is engineered using KiCad, which provides the schematic-capture, PCB-layout, and design-verification environment. The controller platform integrated into the design is an Arduino UNO microcontroller.

2. Power Supply and Regulation Architecture

The board accepts a 6–8 V battery input as its primary power source. From this input, the design provides separate regulated supply paths for the low-power control and sensing electronics and for the servo actuators.

Subsystem	Specification	Design role
PCB type	2-layer PCB	Front signal routing and back-layer copper pour.
Design software	KiCad	Schematic capture, PCB layout, and verification.
Controller	Arduino UNO	Main microcontroller platform.
Power input	6–8 V battery	Primary input source.
Logic supply	Regulated external 5 V	Supplies core controller and sensor logic.
Servo supply	Separate regulated servo supply	Isolates high-current actuator loads.

Subsystem	Specification	Design role
Grounding	Common GND	Provides a unified electrical reference.

The regulated external 5 V rail is intended for the Arduino UNO and associated logic or sensor circuitry. The separate servo rail prevents the large and rapidly changing current demand of the actuators from directly disturbing the main logic supply. A common GND architecture connects the supply and signal reference points into one unified return network.

3. Design Verification and DRC Results

The PCB design was verified using the KiCad Design Rules Checker (DRC). The recorded results show no reported design-rule violations, no unconnected items, and no schematic parity errors.

Verification item	Recorded result	Interpretation
PCB verification tool	KiCad DRC	Design Rules Checker used for PCB verification.
DRC violations	0 errors detected	No reported KiCad design-rule violations.
Unconnected items	0	No unrouted or unconnected items reported.
Schematic parity errors	0	Schematic and PCB layout are synchronized according to the parity check.

A zero DRC-violation result indicates that the KiCad checker did not identify errors against the configured design rules. Zero unconnected items indicates that the intended electrical nets are represented as connected in the project. Zero schematic parity errors indicates that the schematic and physical PCB layout are synchronized according to the performed comparison.

These results establish design consistency within the KiCad project. They do not independently verify fabricated-board performance, power-supply accuracy, thermal behavior, battery safety, signal integrity under load, solder quality, or correct operation of the assembled product. Those properties require physical inspection and measurement.

4. Recommended Next Checks

- Confirm battery polarity and verify that the 6–8 V input remains within the intended operating range.
- Measure the regulated external 5 V logic rail before connecting the Arduino UNO and sensors.
- Verify the separate servo supply voltage and peak-current capability.
- Inspect common-GND continuity and connector orientation.
- Perform a fabricated-PCB short-circuit and continuity inspection.
- Measure voltage sag, current, temperature, and controller stability during servo operation.

5. Conclusion

The AquaSense PCB is a two-layer KiCad design integrating an Arduino UNO controller, a 6–8 V battery input, a regulated external 5 V logic rail, a separate regulated servo supply, and common grounding. KiCad verification records show 0 DRC violations, 0 unconnected items, and 0 schematic parity errors. The design is suitable for controlled fabrication or assembly review, followed by physical electrical, thermal, safety, and functional validation.